Lovász Local Lemma (LLL)

- "bad" events $A_1, \dots, A_n$

  wts possible that <u>none</u> occur.

  i.e. want $\mathbb{P}\left(\bigcap_{i=1}^{n} A_i^c\right) > 0$.

- <u>indep</u> if $A_1, \dots, A_n$ indep.

  then $\mathbb{P}\left(\bigcap_{i=1}^{n} A_i^c\right) = \prod \mathbb{P}(A_i^c) = \prod (1 - \mathbb{P}(A_i)) > 0$.

  as long as no bad event is certain.

- <u>union</u> if $\sum_{i=1}^{n} \mathbb{P}(A_i) < 1$ then $\mathbb{P}\left(\bigcap_{i=1}^{n} A_i^c\right) = 1 - \mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right)$

  $\leq \sum_{i=1}^{n} \mathbb{P}(A_i) < 1$ (union bound)

LLL tool for the case $A_1, \dots, A_n$ "nearly indep"

i.e. each A_i only depends on few other A_j's, indep from rest.

<u>Def^n</u> event A <u>indep. from a family of events</u> $(B_1, \dots, B_k)$

if $\forall S, T \subseteq [k]$ with S, T disjoint

$$\mathbb{P}\left(A \,\middle|\, \left(\bigcap_{i \in S} B_i\right) \cap \left(\bigcap_{j \in T} B_j^c\right)\right) = \mathbb{P}(A).$$

(warning) not the same as pairwise-indep. !!

Example flip two coins (indep, prob 1/2 / 1/2 for H/T).

$A_{H\cdot}$ = first is heads $\{HH, HT\}$

$A_{\cdot H}$ = second " " $\{HH, TH\}$

A_{same} = both coins same $\{HH, TT\}$.

then $A_{H\cdot} \perp A_{\cdot H}$, $A_{H\cdot} \perp A_{same}$, $A_{\cdot H} \perp A_{same}$

$\mathbb{P}(A_{H\cdot} \cap A_{same}) = \mathbb{P}(\{HH\}) = 1/4$.

$\mathbb{P}(A_{H\cdot}) \cdot \mathbb{P}(A_{same}) = \mathbb{P}(\{HH, HT\}) \cdot \mathbb{P}(\{HH, TT\}) = 1/4$

but A_{same} not indep from (family) $A_{H\cdot}, A_{\cdot H}$

$\mathbb{P}(A_{same} \mid A_{H\cdot} \cap A_{\cdot H})$ = (take $S = \{1,2\}$)



directed graph $D = ([n], E)$

Def^n A digraph is called a dependency digraph for events $A_1, \dots, A_n$ if for every i;

→ event A_i is indep of family $\{A_j : j \neq i, i \not\to j\}$.

(i.e. $\{j_1, \dots, j_k\}$ non-nbhrs of i then require that A_i indep of $\{A_{j_1}, \dots, A_{j_k}\}$.

↑ not nec. unique.

LLL (gen. form).

Let D be a dep. digraph for events $A_1, \dots, A_n$.

Supp. $x_1, \dots, x_n \in [0,1)$ satisfy

$$\mathbb{P}(A_i) \le x_i \prod_{j : i \to j} (1 - x_j) \qquad \forall i.$$

Then $$\mathbb{P}\left(\bigcap_{i=1}^n A_i^c\right) \ge \prod_{i=1}^n (1 - x_i) > 0$$

Proof

Claim For any $S \subsetneq [n]$, any $i \notin S$,

$$\mathbb{P}\left(A_i^c \mid \bigcap_{j \in S} A_j^c\right) \ge 1 - x_i.$$

$S = \emptyset$, $\bigcap_{j \in \emptyset} A_j^c = \Omega$, $\mathbb{P}(A_i^c \mid \Omega) = \mathbb{P}(A_i^c) \ge 1 - x_i$

Then claim $\Rightarrow$ thm by 'telescoping products'

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i^c\right) = \underbrace{\mathbb{P}(A_1^c)}_{\ge 1-x_1} \cdot \underbrace{\mathbb{P}(A_2^c \mid A_1^c)}_{\ge 1-x_2} \cdot \underbrace{\mathbb{P}(A_3^c \mid A_1^c \cap A_2^c)}_{\ge 1-x_3} \cdot \dots$$

$$\dots \cdot \underbrace{\mathbb{P}(A_n^c \mid A_1^c \cap \dots \cap A_{n-1}^c)}_{\ge 1-x_n}$$

$$\ge \prod_{i=1}^n (1 - x_i).$$

Claim (repeat) for any $S \subsetneq [n]$, any $i \notin S$,

$$\mathbb{P}\left(A_i^c \mid \bigcap_{j \in S} A_j^c\right) \geq 1 - x_i.$$

Pf of claim → proceed by induction on s.

base $|S| = 0$ $\quad \mathbb{P}\left(A_i^c \mid \bigcap_{j \in \emptyset} A_j^c\right) = \mathbb{P}(A_i^c) = 1 - \mathbb{P}(A_i)$ (deph)

dependency digraph.

$$\leq x_i \prod_{j : i \to j} (1 - x_j) \qquad \forall i$$

inductive step assume claim $\forall S : |S| \leq r-1$.

take $|S| = r$, $i \notin S$. Let $S_1 = \{j \in S : i \to j\}$ $= \{s_1, s_2, \ldots, s_a\}$, $a \leq r$. "dep. part"

$S_0 = \{j \in S : i \not\to j\}$ "indep part".

w.t.s $\quad \mathbb{P}\left(A_i \mid \underbrace{\left(\bigcap_{j \in S_1} A_j^c\right)}_{=: B} \cap \underbrace{\left(\bigcap_{j \in S_0} A_j^c\right)}_{=: C}\right) \leq x_i$

$$\mathbb{P}(A_i \mid B \cap C) = \frac{\mathbb{P}(A_i \cap B \mid C)}{\mathbb{P}(B \mid C)} \leq \frac{\mathbb{P}(A_i \mid C)}{\mathbb{P}(B \mid C)} = \frac{\mathbb{P}(A_i)}{\mathbb{P}(B \mid C)} \qquad (1)$$

(*) $\neq 0$

$$\mathbb{P}(A_i) \leq x_i \prod_{j : i \to j} (1 - x_j) \qquad (2).$$

Note if $S_1 = \emptyset$ then $\mathbb{P}\left(A_i \mid \bigcap_{j \in S_0} A_j^c\right) = \mathbb{P}(A_i) \leq x_i$ ∴ done, so may assume $|S_0| \leq r-1$..

Proof of claim (ct'd).

("dep" ↓ "indep" ↓)

$$\mathbb{P}(B \mid C) = \mathbb{P}\left(A_{S_1}^c \cap A_{S_2}^c \cap \dots \cap A_{S_a}^c \mid C\right)$$

$$= \mathbb{P}(A_{S_1}^c \mid C) \cdot \mathbb{P}(A_{S_2}^c \mid A_{S_1}^c \cap C) \cdot$$

($\mathbb{P}(A_{S_1}^c \mid C)$: $\le r-1$ events; $\mathbb{P}(A_{S_2}^c \mid A_{S_1}^c \cap C)$: $\le r-1$)

$$\dots \cdot \mathbb{P}(A_{S_a}^c \mid A_{S_1}^c \cap \dots \cap A_{S_{a-1}}^c \cap C)$$

($\le r-1$)

by IH. ↓

$$\geq (1-x_{S_1})(1-x_{S_2})\cdots(1-x_{S_a}) = \prod_{j : i \to j} (1-x_j).$$

Hence with (1), (2).

$$\mathbb{P}(A_i \mid B \cap C) \overset{(1)}{\leq} \frac{\mathbb{P}(A_i)}{\mathbb{P}(B \mid C)} \leq \frac{x_i \prod_{j : i \to j} (1-x_j)}{\prod_{j : i \to j} (1-x_j)} = x_i$$

□.

Rem. $\mathbb{P}(A \mid B)$ not defined if $\mathbb{P}(B)$

fix: read $\mathbb{P}(A \mid B) \geq t$ as shorthand for ..

$\mathbb{P}(A \cap B) \geq t \cdot \mathbb{P}(B)$. (then ok).

+ for def^n of dep. digraph read $\mathbb{P}(A \mid B) = \mathbb{P}(A)$

as $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$ ← ok for $\mathbb{P}(B) = 0$.